Wireless InSite®

3D Wireless Communication and Propagation Prediction Software

INSTALLATION GUIDE



REMCOM®
Electromagnetic Simulation Software

Wireless InSite® Installation Guide

Version 3.4.4

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Chapter 1

Installation

In this document, you will learn...

- → the system requirements for an Wireless InSite® installation
- → how to install Wireless InSite on a variety of platforms
- → how to license your new software
- → how to diagnose problems when running the software
- → how to obtain additional information about Wireless InSite

1.1 Introduction

Below are the main steps to installing and licensing Wireless InSite 3.4.4 :

1. Install the Wireless InSite application
2. Obtain a license file from Remcom
3. Place the license file in its proper directory
4. Load the license file

In addition you may:

1. Use a floating license server to validate licensing for multiple machines on a network
2. Validate the license using a transportable USB hardware key

1.2 System Requirements

Wireless InSite is supported on the following operating systems:

- Microsoft Windows 7, 8, and 10, 64-bit editions
- Red Hat Enterprise Linux 6, 64-bit edition (support is for the calculation engine only)

Hardware requirements:

- 8GB RAM minimum, 16GB (or more) recommended.
- 1024 x 768 display resolution (1280 x 1024 or higher preferred).
- Graphics card with CUDA compute capability between 5 and 8.6.[1]
- Most Recent NVIDIA video driver.
- 550 MB free space on hard drive for program files.

Optional:

- MatLab Component Runtime v9.0; see 1.3.1. This is required for the BER calculation.
- FlexLM dongle driver if you were issued a USB hardware licensing dongle; see 2.2.1

1.3 Installing Wireless InSite

The installation files are provided on the Wireless InSite installation CD and are also obtainable via web download. The installation process installs both the application and its supporting files.

1.3.1 Installing on Windows

Log on as an administrator to install the Wireless InSite software.

If installing from CD, insert the 64-bit CD. Run `setup.exe` on the CD if it does not start automatically. If installing from a file download, run the 64-bit setup file:`WirelessInSite.x.x.x.x.Win.x64.exe`. Follow the on-screen instructions to complete the Wireless InSite installation.

If you want to run the Bit-error rate calculation provided in the Communication System tool box you will also need to install the MatLab Component Runtime. The installer is called `MCR_R2016a_win64_installer.exe` and is also included with the provided files.

When the installation process has finished, proceed to chapter 2 to license your software.

[1] GPU must be a CUDA capable card and compatible with NVIDIA OptiX 6.0 Based on the release notes for OptiX 6.0 and the timing of its release, CUDA capable cards with compute capability between 5 and 8.6 should be supported. Contact NVIDIA support for specific recommendations.

1.3.2 Installing on Linux

A Linux version of the calculation engine is provided. The prerequisites for installing Wireless InSite on a Linux system are:

- `uudecode` must be installed. Without uudecode, the installation script will print a message and exit. On RHEL6, uudecode is provided by the sharutils package. One way to install this package on RHEL6 is to use the command (as root) `yum install sharutils`.

To install it:

1. Log on as a user with appropriate write permissions for the desired destination directory.
2. Run the installer with the destination directory as a parameter.
3. Accept the terms of the EULA.

   ✓ If you have another Remcom, Inc. application installed you might be asked to overwrite pre-existing files. In this case select the *[N]one* option.
4. At this point a `remcom` directory will be created and organized as follows:

   remcom - Directory of all installed Remcom products.

   bin - Supporting scripts for the `remcom-shell` script in each versions `bin` directory.

   flexlm - License libraries required to run (`calcprop` or `wibatch`).

   licenses - Location to store license files for flexlm to read.

   Wireless InSite - All versions of Wireless InSite are stored here.

   3.4.4 /architecture - The installed version of Wireless InSite and the architectural platform of the computer that the calculation engine (`calcprop` or `wibatch`) will run on.

   bin - Location to launch the *remcom-shell* script and run the calculation engine.

   doc - Supporting documentation for version 3.4.4 of Wireless InSite.

Supported Linux architectures are:

- Linux-x86_64RHEL6 (64-bit platform)

To launch the calculation engine, go to the correct architecture directory and run the `remcom-shell` script which ensures that the license server is functioning correctly before running the calculation engine. At this point type (`calcprop` or `wibatch`) with the required parameters as described in the manual to run the calculation.

Chapter 2

Licensing Wireless InSite

This software is licensed using Flexera Software's FLEXlm system, which supports file-based (including DEMO licenses) and server-based licensing schemes.

2.1 DEMO License Installation

To install a DEMO license, save the license file provided by Remcom to a location in your filesystem. For example, on Windows a good location might be `C:\Program Files\Common Files\Remcom\licenses`, which is the default location for Remcom license files. When starting Wireless InSite, the Choose License dialog will appear (see Figure 2.1). Click the ellipses button after the *License location* field and browse to the license file, select it and click "Open" then "OK". The "OK" button in the Choose License dialog should then enable, indicating that Wireless InSite is licensed. Click "OK" to begin using Wireless InSite.

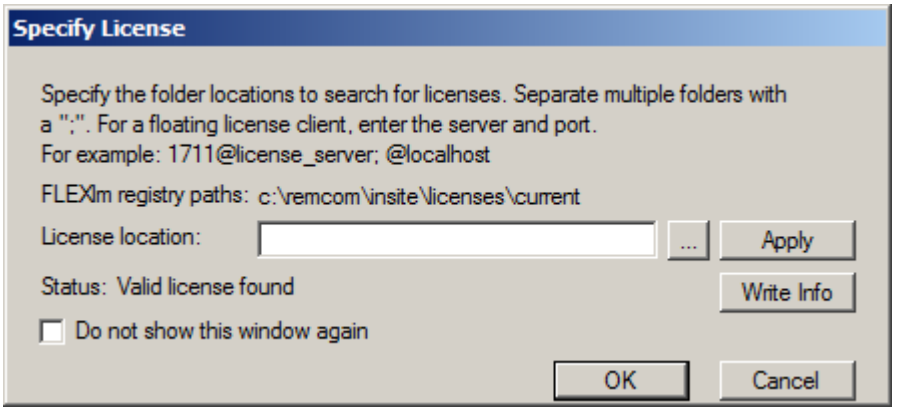


Figure 2.1: Choose License dialog

2.2 Non-DEMO License Installation

For non-DEMO license installations, there are two basic licensing configurations: file-based (with or without a FlexID USB key) and server-based. All license configuration types require a license file (`*.lic`) that is provided by Remcom.

A file-based license configuration is the most common setup. It is a "node-locked" license, meaning that it can be used only on the computer for which you provided Remcom a HOSTID, or only with the USB key supplied with your license (if applicable).

Server-based configurations are used primarily for managing floating licenses, but are sometimes required for special situations where license checkout counts must be enforced. Which type of license configuration you are using will be specified when you obtain your license.

2.2.1 USB Key Installation

If you were not supplied a FlexID USB key, proceed to Section 2.3 or Section 2.4, whichever is appropriate for the type of license configuration you have purchased.

If you were supplied a FlexID USB key, both the license file that was sent to you (`*.lic`) and the USB key are required to run Wireless InSite.

The FlexID USB key that you received will be either a FlexID-9 (green) or FlexID-10 (blue). Driver software must be installed on your computer for the keys to be recognized by the licensing software. The first thing you must do, if you have not previously done it, is install the software for the USB key.



Figure 2.2: FlexID-9 (green) and FlexID-10 (blue) USB Keys

The USB key software can be downloaded from: http://www.modularmerchant.com/clients/remcom/download using the password "EQMK4DJU9JD8EJ7SZ5ZV" (without the quotes).

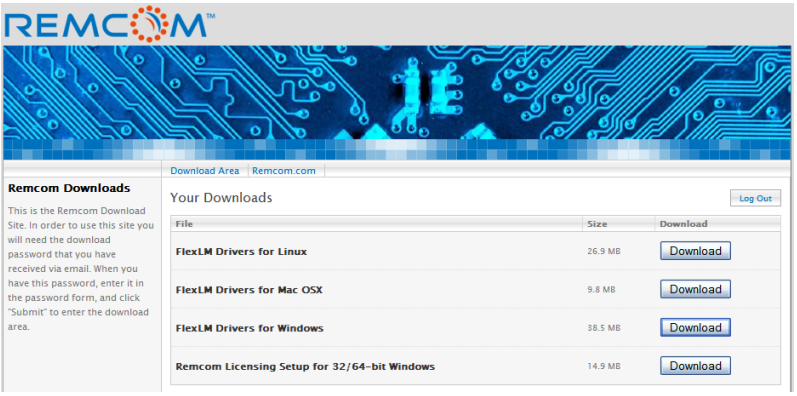


Figure 2.3: Downloading License Support Files

Follow the instructions in the following subsection appropriate for your platform for installing the USB key drivers. After the FlexID driver setup is complete, you may wish reboot your computer to ensure that the installer's changes take effect. Then insert the USB key into a free USB port on your computer and continue on to Section 2.3.

2.2.2 Windows USB Key Driver Installation

For Windows, download "FlexID Dongle Driver Installers for Windows" unzip it, and run "FlexID_Dongle_Driver_Files/Windows/FLEXId_Dongle_Driver_Installer-v16.0.0.exe". This package can install drivers for both FlexID-9 (green) and FlexID-10 (blue) keys. Select "FLEXId 9 Drivers" or "FLEXId10 10 Drivers" depending upon which key you have and finish the installation.

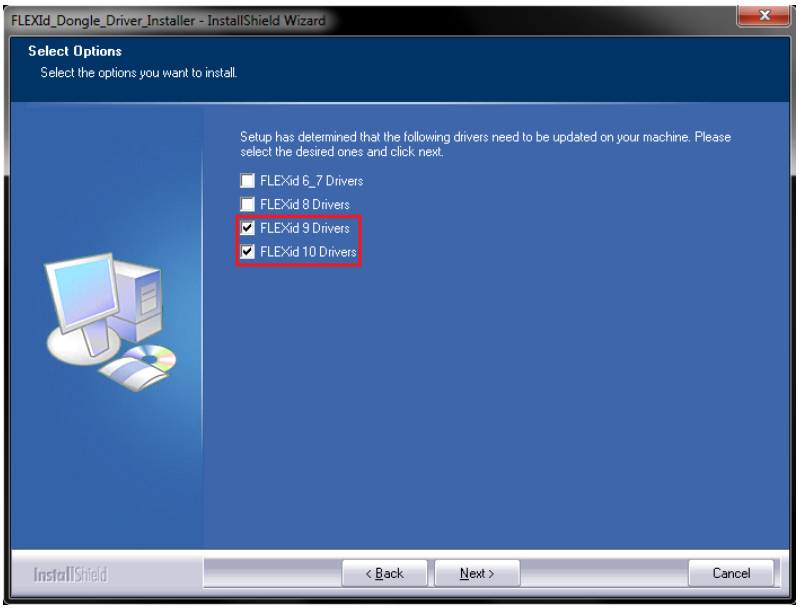


Figure 2.4: FLEXId Driver Installation

2.2.3 Linux USB Key Driver Installation

For Linux, download "FlexID Dongle Driver Installers for Linux", unzip it and install the RPM appropriate for your platform (RedHat or SUSE) and USB key (FlexID-9, green or FlexID-10, blue). This is typically done using the RPM command as the root user, e.g. `rpm -ivh FlexID_Dongle_Driver_Files/RedHat-FlexID-10/WkRt-Lin64-6.0.501.1.x86_64.rpm`.

2.3 File-based Configuration

In this configuration, the license file provided to you by Remcom technical support or your sales representative is required. If you were supplied a FlexID USB key, both the license file and the USB key are required to run Wireless InSite. The instructions in Section 2.2.1 should be followed before continuing here.

2.3.1 License File Specification

Remcom technical support or your sales representative will provide you with your Wireless InSitelicense (`*.lic`) file. Save your license file to a location on your hard drive. For example, on Windows a convenient location might be `C:\Program Files\Common Files\Remcom\licenses`. On Linux, a convenient location might be the `remcom/licenses` directory that is created during installation.

To specify the location of a license file:

- On Windows, the Choose License dialog will appear when you open Wireless InSite (see Figure 2.1). Either enter the folder where you saved your license file, or click the ellipses button after the *License*

Location field, browse to the license file, select it and click “Open” then “Apply”. The *status* should update to say *Valid license found*, indicating that Wireless InSite is licensed. Click “OK” to begin using Wireless InSite.

- On Linux, create an environment variable while in the `remcom-shell` using the command “export REMCOMINC_LICENSE_FILE={license_file_location}”, where “{license_file_location}” is substituted with the path to the desired `*.lic` file.

Note that if you elect to not set the REMCOMINC_LICENSE_FILE environment variable, the calculation engines may still be run via the command line using the appropriate arguments. See Chapter 25.1 of your Wireless InSiteReference Manual for more information.

2.4 Server-based Configuration

License server-based configurations require that a license server be installed, configured and running and for the user to specify the license server location when starting Wireless InSite.

If you are not responsible for configuring the license server, obtain the License Server Location you are to use from the license server manager. This will be of the form “port@servername”; e.g. “1711@license_server”. Then skip to “Specify License Server Location” below, where you will enter this information in Wireless InSite’s Choose License dialog.

If you were supplied a FlexID USB key with your server-based license file, both the license file and the USB key are required to license Wireless InSite. The instructions in Section 2.2.1 should be followed before continuing here.

2.4.1 Installing the License Server - Windows

To configure server-based licensing on Windows, you must first install the server software if you have not already done so. Do this by downloading and installing the “Remcom Licensing Setup for 32/64-bit Windows” package from `http://www.modularmerchant.com/clients/remcom/download` using the password “`EQMK4DJU9JD8EJ7SZ5ZV`” (without the quotes).

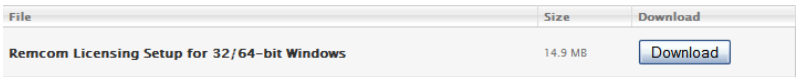


Figure 2.5: Downloading the Windows License Server Software

Please note that you must have Administrator or Power User privileges on your computer when installing the license server software.

Remcom technical support or your sales representative will have provided you with a license file (`*.lic`) for use with the license server. Open the license folder by selecting **Start ► All Programs ► Remcom ► Licensing ► Open License Folder**. Save the license file to that folder (which is `C:\Program Files\Common Files\Remcom\licenses`).

View the file and if the “SERVER” line contains “enterhostnamehere”, change it to the actual hostname of the server. Optionally, specify a specific port for the license server to listen at by adding the port number at

the end of the “SERVER” line. By default, a port number from 25000 to 25005 will be used if a port number is not specified. Save the license file if changes were made.

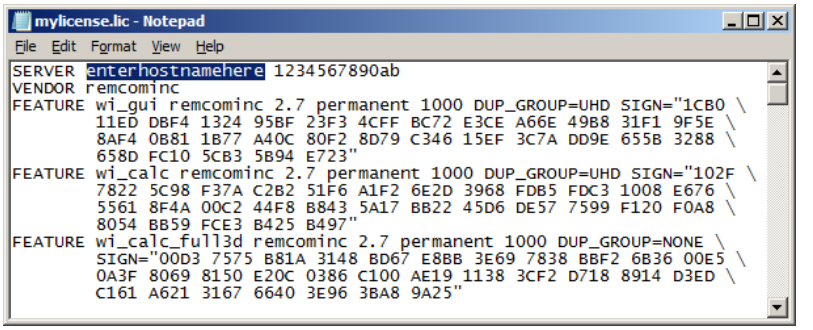


Figure 2.6: Editing the license file hostname parameter

Optionally, specify a specific port for the license server to run at by adding the port number at the end of the “SERVER” line. By default, Windows picks a port number from 25000 to 25005. When complete, save the license file. Note: If the file already has the correct hostname, no changes are required.

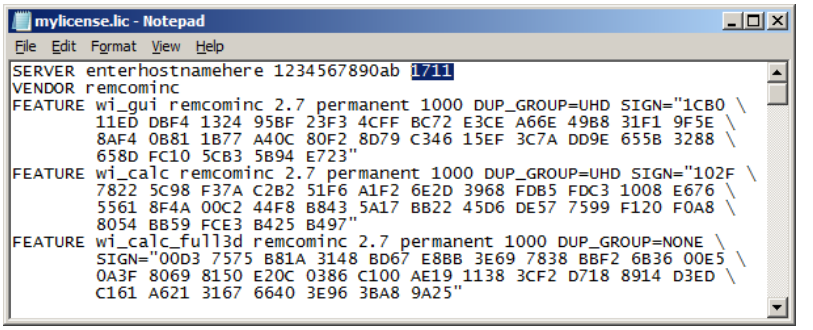


Figure 2.7: Specifying the license server port

Select **Start ► All Programs ► Remcom ► Licensing ► Add License Service**. This action create the “Remcom License Sever” Windows service and requires Administrator or Power User privileges to work.

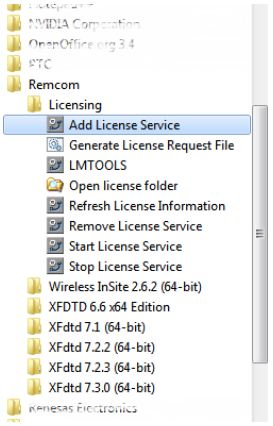


Figure 2.8: Adding the License Server Service

Next, select **Start ► All Programs ► Remcom ► Licensing ► LMTOOLS** to open the server configuration tool. Navigate to the “Start/Stop/Reread” tab, click the “Start Server button” to start the server and close LMTOOLS. Reboot your computer to ensure that the Remcom License Server installation is successful.

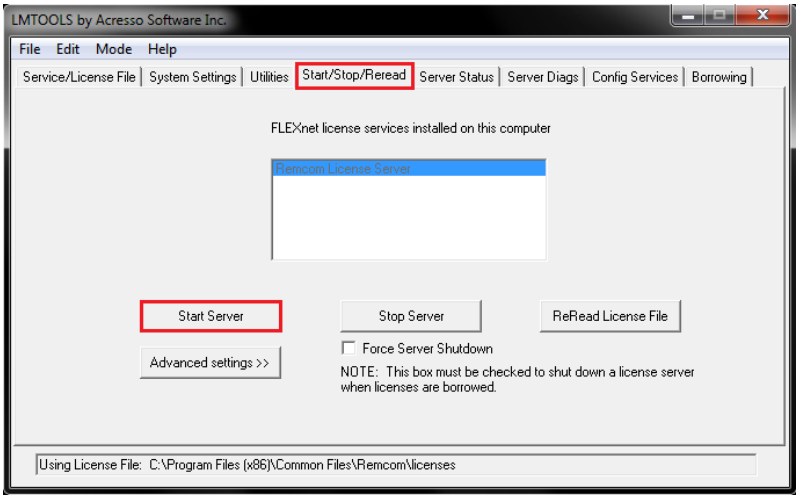


Figure 2.9: Starting the License Server

The license server software installation is now complete. If you did not specify a port number on the “SERVER” line of the license file, the license server location to be specified in the application is “@hostname” where “hostname” is the name you entered above to replace “enterservernamehere”. Otherwise, the license server location is “port@hostname” where “port” and “hostname” are the port number and host name specified in the license file. With this information, continue below at Section 2.4.3, “Specify License Server Location” to license Wireless InSite.

2.4.2 Installing the License Server - Linux

If you are not responsible for configuring the license server, skip to Section 2.4.3, "Specify License Server Location" below.

The license server software for Linux is packaged with Wireless InSite and can be found in the `remcom/flexlm/v11.10.1/{platform}` directory after installation, where {platform} is `i86_lsb` for 32-bit servers and `x64_lsb` for 64-bit servers. There are several executables available:

- `lmgrd`: The license server daemon
- `remcominc`: Remcom's vendor damon, used by `lmgrd`
- `lmutil`: License server utility (query, stop, reread, etc.)

Remcom technical support or your sales representative will have provided you with a license file (`*.lic`) for use with the license server. Save the license file to a convenient location, such as the `remcom/licenses` directory that is created during installation. Before using the license, the license file must be edited to insert the appropriate server name.

View the file and if the "SERVER" line contains "enterservernamehere", change it to the actual hostname of the server. Optionally, specify a specific port for the license server to listen at by adding the port number at the end of the "SERVER" line. By default, a port number from 25000 to 25005 will be used if a port number is not specified. Save the license file if changes were made.

Run the license server using the command {`install_dir`}`/remcom/flexlm/v11.10.1/`{`platform`}`/lmgrd -c` {`license_file`} where items between curly braces are substituted with appropriate values. If no error messages are printed out, the license server is ready.

If you did not specify a port number on the "SERVER" line of the license file, the license server location to be specified in the application is "@hostname" where "hostname" is the name you entered above to replace "enterservernamehere". Otherwise, the license server location is "port@hostname" where "port" and "hostname" are the port number and host name specified in the license file. With this information, continue below at Section 2.4.3, "Specify License Server Location" to license Wireless InSite.

Note that Wireless InSite does not require an individual license server. Since it has its own daemon, it can run alongside other engineering packages like Matlab, Pro/E, etc. Additionally, if you already have `lmgrd` running on your system, you can have that daemon serve Wireless InSite licenses as well.

2.4.3 Specify License Server Location

If you are using a floating license and it is managed by an IT department, that department will provide you with the license server location you will enter in the Choose License dialog of Wireless InSite on Windows or set as the value of the `REMCOMINC_LICENSE_FILE` environment variable on Linux. The server location is of the format "port@servername". For example, "1711@license_server". When you start Wireless InSite on Windows, the Choose License dialog will appear (see Figure 2.1). Enter the server location in *License Location* then click "Apply". The *status* should update to say *Valid license found*, indicating that Wireless InSite is licensed. Click "OK" to begin using Wireless InSite. On Linux, create an environment variable while in the `remcom-shell` using the command "export REMCOMINC_LICENSE_FILE={port@servername}"

Chapter 3

Troubleshooting

3.1 'Valid license could not be found'

If, after following the instructions in chapter 2, Wireless InSite cannot find a valid license, ensure that:

- None of the *FEATURE* lines in the license file received from Remcom have been edited and the *SERVER* line has not had its hostid field modified.
- The LICENSE LOCATION of the Choose License dialog correctly specifies the appropriate server specification.

The WRITE INFO button in the Choose License dialog provides extended information about the licensing process, and may contain error messages that help in determining why a license cannot be obtained. Review this information to try and identify the problem. If it persists send this information to Remcom technical support at `support@remcom.com`.

3.2 'Wireless InSite calculation engine license is not available'

If, after following the instructions in chapter 2, `calcprop` cannot find a valid license open a command window and type the following commands:

- `cd installation_directory/calc??` where ?? is the bit architecture of the version you are trying to run.
- `set REMCOMINC_LICENSE_FILE=@localhost`.
- `calcprop --project=project_setup_file` where `project_setup_file` is the absolute path to the *.setup file that you were trying to run. At this point you should see normal output from `calcprop` indicating the status of the calculation.

3.3 Unable to run the BER calculation.

If you see this message when running a Communications System job then Wireless InSite is having problems interacting with the MatLab Component Runtime (MCR). There are several reasons that this may occur:

- You still need to install the MCR as detailed in 1.3.
- There is an issue with Windows user account permissions. In this case you can try one of the following procedures:
  - Right-click on the Wireless InSite shortcut in the Start menu and select *Run as Administrator* from the context menu. If this is not available, then select the same option from the context menu of the Start menu shortcut *Programs→Accessories→Command Prompt*. In the command prompt window that appears go to the installation directory and launch `InSite.exe`.
  - Modify the installation procedure:
    1. Install the MCR with admin priviledges.
    2. Copy the installation to another user accessable directory.
    3. Add the new location manually to the PATH environment variable for the user; not the system PATH variable.

    * An administrator could install once to a USB storage drive and skip step 1 when updating multiple computers.

3.4 Unable to save project to API document

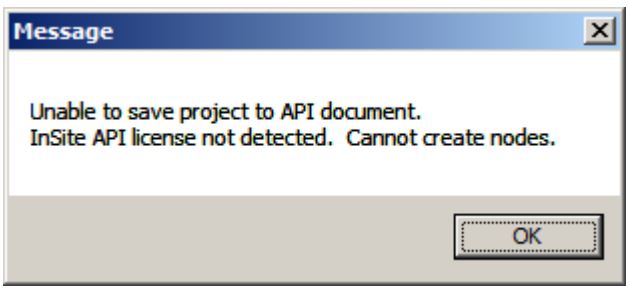


Figure 3.1: Error when saving a project in order to run a model.

This may happen when a new license location is specified in the *License location* of the SPECIFY LICENSE window. The application will need to be restarted to correct this issue.

3.5 High Definition Screen - Matlab 9.0 conflict

A problem can arise when running on Windows with Matlab 9.0 installed, when users have a high definition (HD) screen. Windows High DPI scaling is set to $> 100\%$. (For people with large resolution displays this

is common; in fact Windows may do this by default when setting up the display.) When clicking on 'About' in the user interface, Wireless InSite resizes and many of the buttons become inaccessible. This appears to be due to a bug inside Matlab's DLL (See: https://www.mathworks.com/support/bugreports/1054442).

A workaround for this is to do the following:

- Find InSite.exe
- Right-click and go to Properties/Compatibility
- Select the "Change high DPI settings" button
- And select "Override high DPI scaling behavior" and set the drop down to "System (Enhanced)"

Chapter 4

Open Source Licenses

Open source software was used in the development of Wireless InSite. The license agreements for all of these modules are listed below.

4.1 Open Asset Import Library

```
Open Asset Import Library (assimp)

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******************************************************************************
```

```
AN EXCEPTION applies to all files in the ./test/models-nonbsd folder.
These are 3d models for testing purposes, from various free sources
on the internet. They are - unless otherwise stated - copyright of
their respective creators, which may impose additional requirements
on the use of their work. For any of these models, see
<model-name>.source.txt for more legal information. Contact us if you
are a copyright holder and believe that we credited you inproperly or
if you don't want your files to appear in the repository.


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```

4.2 NVIDIA CUDA Toolkit

```
1. NVIDIA Software License Agreement
-------------------------------------


Release Date: October 20, 2016
------------------------------


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-------------------------------------------------------

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1.8. Term and Termination

1.8.1. AGREEMENT, Licenses and Services

This SLA shall become effective upon the Effective Date, each Supplement upon their acceptance, and both this SLA and Supplements shall continue in effect until your last access or use of the Licensed Software and/or services hereunder, unless earlier terminated as provided in this Term and Termination section. Each Licensed Software license ends at the earlier of (a) the expiration of the applicable license term, or (b) termination of such license or the AGREEMENT. Each service ends at the earlier of (x) the expiration of the applicable service term, (y) termination of such service or the AGREEMENT, or (z) expiration or termination of the associated license and no credit or refund will be provided upon the expiration or termination of the associated license for any service fees paid.

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Neither party will be responsible for any failure or delay in its performance under the AGREEMENT (except for any payment obligations) to the extent due to causes beyond its reasonable control for so long as such force majeure event continues in effect.

The AGREEMENT will be governed by and construed under the laws of the State of Delaware and the United States without regard to the conflicts of law provisions thereof and without regard to the United Nations Convention on Contracts for the International Sale of Goods. The parties consent to the personal jurisdiction of the federal and state courts located in Santa Clara County, California. You acknowledge and agree that a breach of any of your promises or agreements contained in the AGREEMENT may result in irreparable and continuing injury to NVIDIA for which monetary damages may not be an adequate remedy and therefore NVIDIA is entitled to seek injunctive relief as well as such other and further relief as may be appropriate. If any court of competent jurisdiction determines that any provision of the AGREEMENT is illegal, invalid or unenforceable, the remaining provisions will remain in full force and effect. Unless otherwise specified, remedies are cumulative.

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2. AGREEMENT means this SLA and all associated Supplements entered by the parties referencing this SLA.

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4. Confidential Information means the Licensed Software (unless made publicly available by NVIDIA without confidentiality obligations), and any NVIDIA business, marketing, pricing, research and development, know-how, technical, scientific, financial status, proposed new products or other information disclosed by NVIDIA to you which, at the time of disclosure, is designated in writing as confidential or proprietary (or like written designation), or orally identified as confidential or proprietary or is otherwise reasonably identifiable by parties exercising reasonable business judgment, as confidential. Confidential Information does not and will not include information that: (i) is or becomes generally known to the public through no fault of or breach of the AGREEMENT by the receiving party; (ii) is rightfully known by the receiving party at the time of disclosure without an obligation of confidentiality; (iii) is independently developed by the receiving party without use of the disclosing partys Confidential Information; or (iv) is rightfully obtained by the receiving party from a third party without restriction on use or disclosure.

5. Contractor means an individual who works primarily for your Enterprise on a contractor basis from your secure network.

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   more than fifty percent (50%) of the issued and
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   code.

 14.

   Supplement means the additional terms and conditions
   beyond those stated in this SLA that apply to certain
   Licensed Software licensed hereunder.


2. CUDA Supplement to Software License Agreement
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Release date: March 24, 2016
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The terms set forth in this CUDA Supplement (Supplement)
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CUDA family of software products, namely the CUDA Toolkit and
associated drivers, sample source code and the CUDA Nsight
Visual Studio Edition (collectively, the CUDA Licensed
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Component : CUDA Runtime
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  Mac OSX : libcudart.dylib, libcudart_static.a, libcudadevrt.a
  Linux   : libcudart.so, libcudart_static.a, libcudadevrt.a
  Android : libcudart.so, libcudart_static.a, libcudadevrt.a

Component : CUDA FFT Library
  Windows : cufft.dll, cufftw.dll, cufft.lib, cufftw.lib
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  Android : libcufft.so, libcufft_static.a, libcufftw.so, libcufftw_static.a

Component : CUDA BLAS Library
  Windows : cublas.dll, cublas_device.lib
  Mac OSX : libcublas.dylib, libcublas_static.a, libcublas_device.a
  Linux   : libcublas.so, libcublas_static.a, libcublas_device.a
  Android : libcublas.so, libcublas_static.a, libcublas_device.a

Component : NVIDIA "Drop-in" BLAS Library
  Windows : nvblas.dll
  Mac OSX : libnvblas.dylib
  Linux   : libnvblas.so

Component : CUDA Sparse Matrix Library
  Windows : cusparse.dll, cusparse.lib
  Mac OSX : libcusparse.dylib, libcusparse_static.a
  Linux   : libcusparse.so, libcusparse_static.a
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Component : CUDA Linear Solver Library
  Windows : cusolver.dll, cusolver.lib
  Mac OSX : libcusolver.dylib, libcusolver_static.a
  Linux   : libcusolver.so, libcusolver_static.a
  Android : libcusolver.so, libcusolver_static.a

Component : CUDA Random Number Generation Library
  Windows : curand.dll, curand.lib
  Mac OSX : libcurand.dylib, libcurand_static.a
  Linux   : libcurand.so, libcurand_static.a
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Component : CUDA Accelerated Graph Library
  Windows : nvgraph.dll, nvgraph.lib
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Component : NVIDIA Performance Primitives Library
  Windows : nppc.dll, nppc.lib, nppial.dll, nppial.lib, nppicc.dll, nppicc.lib, nppicom.dll, nppicom.lib, nppidei.dll, nppic
  Mac OSX : libnppc.dylib, libnppc_static.a, libnppial.dylib, libnppial_static.a, libnppicc.dylib, libnppicc_static.a, libnp
  Linux   : libnppc.so, libnppc_static.a, libnppial.so, libnppial_static.a, libnppicc.so, libnppicc_static.a, libnppicom.so,
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Component : Internal common library required for statically linking to cuBLAS, cuSPARSE, cuFFT, cuRAND and NPP
  Mac OSX : libculibos.a
  Linux   : libculibos.a

Component : NVIDIA Runtime Compilation Library
  Windows : nvrtc.dll, nvrtc-builtins.dll
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Component : NVIDIA Optimizing Compiler Library
  Windows : nvvm.dll
  Mac OSX : libnvvm.dylib
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Component : NVIDIA Common Device Math Functions Library
  Windows : libdevice.10.bc
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Component : CUDA Occupancy Calculation Header Library
  All     : cuda_occupancy.h

Component : CUDA Half Precision Headers
  All     : cuda_fp16.h, cuda_fp16.hpp

Component : CUDA Profiling Tools Interface (CUPTI) Library
  Windows : cupti.dll
  Mac OSX : libcupti.dylib
  Linux   : libcupti.so

Component : NVIDIA Tools Extension Library
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  Mac OSX : libnvToolsExt.dylib
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gdal/frmts/gtiff/tif_float.c
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4.3.1 proj4

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4.9 mpeg

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4.12 pthreads

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RELEASE 3.0.0
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(2018-08-08)

General
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Note that this is a new major release. The major version increment
introduces two ABI changes along with other naming changes that will
require recompilation of linking applications and possibly some textual
changes to compile-time macro references in configuration and source
files, e.g. PTW32_* changes to __PTW32_*, ptw32_* to __ptw32_*, etc.

License Change
--------------
With the agreement of all substantial relevant contributors Pthreads4w
version 3, with the exception of four files, is being released under the
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and LGPLv3 licenses and therefore this code may continue to be legally
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reorganise code for aesthetic or practical purposes, or improve build
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likelyhood that not all contributors would be contactable. All
contributors are listed in the file CONTRIBUTORS.

The four files that will remain LGPL but change to v3 are files used to
configure the GNU environment builds:

  aclocal.m4
  configure.ac
  GNUmakefile.in
  tests/GNUmakefile.in

Contributors who have either requested this change or agreed to it when
consulted are:

John Bossom
Alexander Terekhov
Vladimir Kliatchko
Ross Johnson

Pthreads4w version 2 releases will remain LGPL but version 2.11 and later
will be released under v3 of that license so that any additions to
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```

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```
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  We protect your rights with a two-step method: (1) we copyright the
library, and (2) we offer you this license, which gives you legal
permission to copy, distribute and/or modify the library.

  To protect each distributor, we want to make it very clear that
there is no warranty for the free library.  Also, if the library is
modified by someone else and passed on, the recipients should know
that what they have is not the original version, so that the original
author's reputation will not be affected by problems that might be
introduced by others.

  Finally, software patents pose a constant threat to the existence of
any free program.  We wish to make sure that a company cannot
effectively restrict the users of a free program by obtaining a
restrictive license from a patent holder.  Therefore, we insist that
any patent license obtained for a version of the library must be
consistent with the full freedom of use specified in this license.

  Most GNU software, including some libraries, is covered by the
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ordinary GNU General Public License.  This license, the GNU Lesser
General Public License, applies to certain designated libraries, and
is quite different from the ordinary General Public License.  We use
this license for certain libraries in order to permit linking those
libraries into non-free programs.

  When a program is linked with a library, whether statically or using
a shared library, the combination of the two is legally speaking a
combined work, a derivative of the original library.  The ordinary
General Public License therefore permits such linking only if the
entire combination fits its criteria of freedom.  The Lesser General
Public License permits more lax criteria for linking other code with
the library.

  We call this license the "Lesser" General Public License because it
does Less to protect the user's freedom than the ordinary General
Public License.  It also provides other free software developers Less
of an advantage over competing non-free programs.  These disadvantages
are the reason we use the ordinary General Public License for many
libraries.  However, the Lesser license provides advantages in certain
special circumstances.

  For example, on rare occasions, there may be a special need to
encourage the widest possible use of a certain library, so that it becomes
a de-facto standard.  To achieve this, non-free programs must be
allowed to use the library.  A more frequent case is that a free
library does the same job as widely used non-free libraries.  In this
case, there is little to gain by limiting the free library to free
software only, so we use the Lesser General Public License.

  In other cases, permission to use a particular library in non-free
programs enables a greater number of people to use a large body of
free software.  For example, permission to use the GNU C Library in
non-free programs enables many more people to use the whole GNU
operating system, as well as its variant, the GNU/Linux operating
system.

  Although the Lesser General Public License is Less protective of the
users' freedom, it does ensure that the user of a program that is
linked with the Library has the freedom and the wherewithal to run
that program using a modified version of the Library.

  The precise terms and conditions for copying, distribution and
modification follow.  Pay close attention to the difference between a
"work based on the library" and a "work that uses the library".  The
former contains code derived from the library, whereas the latter must
be combined with the library in order to run.

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  0. This License Agreement applies to any software library or other
program which contains a notice placed by the copyright holder or
other authorized party saying it may be distributed under the terms of
this Lesser General Public License (also called "this License").
Each licensee is addressed as "you".

  A "library" means a collection of software functions and/or data
prepared so as to be conveniently linked with application programs
(which use some of those functions and data) to form executables.

  The "Library", below, refers to any such software library or work
which has been distributed under these terms.  A "work based on the
Library" means either the Library or any derivative work under
copyright law: that is to say, a work containing the Library or a
portion of it, either verbatim or with modifications and/or translated
straightforwardly into another language.  (Hereinafter, translation is
included without limitation in the term "modification".)

  "Source code" for a work means the preferred form of the work for
making modifications to it.  For a library, complete source code means
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all the source code for all modules it contains, plus any associated
interface definition files, plus the scripts used to control compilation
and installation of the library.

  Activities other than copying, distribution and modification are not
covered by this License; they are outside its scope.  The act of
running a program using the Library is not restricted, and output from
such a program is covered only if its contents constitute a work based
on the Library (independent of the use of the Library in a tool for
writing it).  Whether that is true depends on what the Library does
and what the program that uses the Library does.

  1. You may copy and distribute verbatim copies of the Library's
complete source code as you receive it, in any medium, provided that
you conspicuously and appropriately publish on each copy an
appropriate copyright notice and disclaimer of warranty; keep intact
all the notices that refer to this License and to the absence of any
warranty; and distribute a copy of this License along with the
Library.

  You may charge a fee for the physical act of transferring a copy,
and you may at your option offer warranty protection in exchange for a
fee.

  2. You may modify your copy or copies of the Library or any portion
of it, thus forming a work based on the Library, and copy and
distribute such modifications or work under the terms of Section 1
above, provided that you also meet all of these conditions:

    a) The modified work must itself be a software library.

    b) You must cause the files modified to carry prominent notices
    stating that you changed the files and the date of any change.

    c) You must cause the whole of the work to be licensed at no
    charge to all third parties under the terms of this License.

    d) If a facility in the modified Library refers to a function or a
    table of data to be supplied by an application program that uses
    the facility, other than as an argument passed when the facility
    is invoked, then you must make a good faith effort to ensure that,
    in the event an application does not supply such function or
    table, the facility still operates, and performs whatever part of
    its purpose remains meaningful.

    (For example, a function in a library to compute square roots has
    a purpose that is entirely well-defined independent of the
    application.  Therefore, Subsection 2d requires that any
    application-supplied function or table used by this function must
    be optional: if the application does not supply it, the square
    root function must still compute square roots.)

These requirements apply to the modified work as a whole.  If
identifiable sections of that work are not derived from the Library,
and can be reasonably considered independent and separate works in
themselves, then this License, and its terms, do not apply to those
sections when you distribute them as separate works.  But when you
distribute the same sections as part of a whole which is a work based
on the Library, the distribution of the whole must be on the terms of
this License, whose permissions for other licensees extend to the
entire whole, and thus to each and every part regardless of who wrote
it.

Thus, it is not the intent of this section to claim rights or contest
your rights to work written entirely by you; rather, the intent is to
exercise the right to control the distribution of derivative or
collective works based on the Library.

In addition, mere aggregation of another work not based on the Library
with the Library (or with a work based on the Library) on a volume of
a storage or distribution medium does not bring the other work under
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the scope of this License.

  3. You may opt to apply the terms of the ordinary GNU General Public
License instead of this License to a given copy of the Library.  To do
this, you must alter all the notices that refer to this License, so
that they refer to the ordinary GNU General Public License, version 2,
instead of to this License.  (If a newer version than version 2 of the
ordinary GNU General Public License has appeared, then you can specify
that version instead if you wish.)  Do not make any other change in
these notices.

  Once this change is made in a given copy, it is irreversible for
that copy, so the ordinary GNU General Public License applies to all
subsequent copies and derivative works made from that copy.

  This option is useful when you wish to copy part of the code of
the Library into a program that is not a library.

  4. You may copy and distribute the Library (or a portion or
derivative of it, under Section 2) in object code or executable form
under the terms of Sections 1 and 2 above provided that you accompany
it with the complete corresponding machine-readable source code, which
must be distributed under the terms of Sections 1 and 2 above on a
medium customarily used for software interchange.

  If distribution of object code is made by offering access to copy
from a designated place, then offering equivalent access to copy the
source code from the same place satisfies the requirement to
distribute the source code, even though third parties are not
compelled to copy the source along with the object code.

  5. A program that contains no derivative of any portion of the
Library, but is designed to work with the Library by being compiled or
linked with it, is called a "work that uses the Library".  Such a
work, in isolation, is not a derivative work of the Library, and
therefore falls outside the scope of this License.

  However, linking a "work that uses the Library" with the Library
creates an executable that is a derivative of the Library (because it
contains portions of the Library), rather than a "work that uses the
library".  The executable is therefore covered by this License.
Section 6 states terms for distribution of such executables.

  When a "work that uses the Library" uses material from a header file
that is part of the Library, the object code for the work may be a
derivative work of the Library even though the source code is not.
Whether this is true is especially significant if the work can be
linked without the Library, or if the work is itself a library.  The
threshold for this to be true is not precisely defined by law.

  If such an object file uses only numerical parameters, data
structure layouts and accessors, and small macros and small inline
functions (ten lines or less in length), then the use of the object
file is unrestricted, regardless of whether it is legally a derivative
work.  (Executables containing this object code plus portions of the
Library will still fall under Section 6.)

  Otherwise, if the work is a derivative of the Library, you may
distribute the object code for the work under the terms of Section 6.
Any executables containing that work also fall under Section 6,
whether or not they are linked directly with the Library itself.

  6. As an exception to the Sections above, you may also combine or
link a "work that uses the Library" with the Library to produce a
work containing portions of the Library, and distribute that work
under terms of your choice, provided that the terms permit
modification of the work for the customer's own use and reverse
engineering for debugging such modifications.

  You must give prominent notice with each copy of the work that the
Library is used in it and that the Library and its use are covered by
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this License.  You must supply a copy of this License.  If the work
during execution displays copyright notices, you must include the
copyright notice for the Library among them, as well as a reference
directing the user to the copy of this License.  Also, you must do one
of these things:

    a) Accompany the work with the complete corresponding
    machine-readable source code for the Library including whatever
    changes were used in the work (which must be distributed under
    Sections 1 and 2 above); and, if the work is an executable linked
    with the Library, with the complete machine-readable "work that
    uses the Library", as object code and/or source code, so that the
    user can modify the Library and then relink to produce a modified
    executable containing the modified Library.  (It is understood
    that the user who changes the contents of definitions files in the
    Library will not necessarily be able to recompile the application
    to use the modified definitions.)

    b) Use a suitable shared library mechanism for linking with the
    Library.  A suitable mechanism is one that (1) uses at run time a
    copy of the library already present on the user's computer system,
    rather than copying library functions into the executable, and (2)
    will operate properly with a modified version of the library, if
    the user installs one, as long as the modified version is
    interface-compatible with the version that the work was made with.

    c) Accompany the work with a written offer, valid for at
    least three years, to give the same user the materials
    specified in Subsection 6a, above, for a charge no more
    than the cost of performing this distribution.

    d) If distribution of the work is made by offering access to copy
    from a designated place, offer equivalent access to copy the above
    specified materials from the same place.

    e) Verify that the user has already received a copy of these
    materials or that you have already sent this user a copy.

  For an executable, the required form of the "work that uses the
Library" must include any data and utility programs needed for
reproducing the executable from it.  However, as a special exception,
the materials to be distributed need not include anything that is
normally distributed (in either source or binary form) with the major
components (compiler, kernel, and so on) of the operating system on
which the executable runs, unless that component itself accompanies
the executable.

  It may happen that this requirement contradicts the license
restrictions of other proprietary libraries that do not normally
accompany the operating system.  Such a contradiction means you cannot
use both them and the Library together in an executable that you
distribute.

  7. You may place library facilities that are a work based on the
Library side-by-side in a single library together with other library
facilities not covered by this License, and distribute such a combined
library, provided that the separate distribution of the work based on
the Library and of the other library facilities is otherwise
permitted, and provided that you do these two things:

    a) Accompany the combined library with a copy of the same work
    based on the Library, uncombined with any other library
    facilities.  This must be distributed under the terms of the
    Sections above.

    b) Give prominent notice with the combined library of the fact
    that part of it is a work based on the Library, and explaining
    where to find the accompanying uncombined form of the same work.

  8. You may not copy, modify, sublicense, link with, or distribute
the Library except as expressly provided under this License.  Any
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attempt otherwise to copy, modify, sublicense, link with, or
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terminated so long as such parties remain in full compliance.

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infringement or for any other reason (not limited to patent issues),
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License and any other pertinent obligations, then as a consequence you
may not distribute the Library at all.  For example, if a patent
license would not permit royalty-free redistribution of the Library by
all those who receive copies directly or indirectly through you, then
the only way you could satisfy both it and this License would be to
refrain entirely from distribution of the Library.

If any portion of this section is held invalid or unenforceable under any
particular circumstance, the balance of the section is intended to apply,
and the section as a whole is intended to apply in other circumstances.

It is not the purpose of this section to induce you to infringe any
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such claims; this section has the sole purpose of protecting the
integrity of the free software distribution system which is
implemented by public license practices.  Many people have made
generous contributions to the wide range of software distributed
through that system in reliance on consistent application of that
system; it is up to the author/donor to decide if he or she is willing
to distribute software through any other system and a licensee cannot
impose that choice.

This section is intended to make thoroughly clear what is believed to
be a consequence of the rest of this License.

  12. If the distribution and/or use of the Library is restricted in
certain countries either by patents or by copyrighted interfaces, the
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an explicit geographical distribution limitation excluding those countries,
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written in the body of this License.

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Such new versions will be similar in spirit to the present version,
but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number.  If the Library
specifies a version number of this License which applies to it and
"any later version", you have the option of following the terms and
conditions either of that version or of any later version published by
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the Free Software Foundation.  If the Library does not specify a
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  14. If you wish to incorporate parts of the Library into other free
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decision will be guided by the two goals of preserving the free status
of all derivatives of our free software and of promoting the sharing
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DAMAGES.

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possible use to the public, we recommend making it free software that
everyone can redistribute and change.  You can do so by permitting
redistribution under these terms (or, alternatively, under the terms of the
ordinary General Public License).

  To apply these terms, attach the following notices to the library.  It is
safest to attach them to the start of each source file to most effectively
convey the exclusion of warranty; and each file should have at least the
"copyright" line and a pointer to where the full notice is found.

    <one line to give the library's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

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Also add information on how to contact you by electronic and paper mail.

You should also get your employer (if you work as a programmer) or your
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school, if any, to sign a "copyright disclaimer" for the library, if
necessary.  Here is a sample; alter the names:

  Yoyodyne, Inc., hereby disclaims all copyright interest in the
  library ‘Frob’ (a library for tweaking knobs) written by James Random Hacker.

  <signature of Ty Coon>, 1 April 1990
  Ty Coon, President of Vice

That’s all there is to it!
```

4.13.1 Qt LGPL Exception

```
Digia Qt LGPL Exception version 1.1

As an additional permission to the GNU Lesser General Public License version
2.1, the object code form of a "work that uses the Library" may incorporate
material from a header file that is part of the Library.  You may distribute
such object code under terms of your choice, provided that:
    (i)   the header files of the Library have not been modified; and
    (ii)  the incorporated material is limited to numerical parameters, data
          structure layouts, accessors, macros, inline functions and
          templates; and
    (iii) you comply with the terms of Section 6 of the GNU Lesser General
          Public License version 2.1.

Moreover, you may apply this exception to a modified version of the Library,
provided that such modification does not involve copying material from the
Library into the modified Library’s header files unless such material is
limited to (i) numerical parameters; (ii) data structure layouts;
(iii) accessors; and (iv) small macros, templates and inline functions of
five lines or less in length.

Furthermore, you are not required to apply this additional permission to a
modified version of the Library.
```

4.14 Shapefile C Library

```
The source for the Shapefile C Library is (c) 1998 Frank Warmerdam, and released under the
following conditions. The intent is that anyone can do anything with the code, but that I do not
assume any liability, nor express any warranty for this code.

As of Shapelib 1.2.6 the core portions of the library are made available under two possible
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Shapelib distribution without explicit copyright license terms (such as this documentation, the
Makefile and so forth) should be considered to have the following licensing terms. Some auxilary
portions of Shapelib, notably some of the components in the contrib directory come under slightly
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I am pleased to receive bug fixes, and improvements for Shapelib. Unless the submissions indicate
otherwise I will assume that changes submitted to me remain under the the above "dual license"
terms. If changes are made to the library with the intention that those changes should be
protected by the LGPL then I should be informed upon submission. Note that I will not generally
incorporate changes into the core of Shapelib that are protected under the LGPL as this would
effectively limit the whole file and distribution to LGPL terms.

Opting for LGPL
For licensee's opting to use Shapelib under LGPL as opposed to the MIT Style license above, and
wishing to redistribute the software based on Shapelib, I would ask that all "dual license"
modules be updated to indicate that only the LGPL (and not the MIT Style license) applies. This
action represents opting for the LGPL, and thereafter LGPL terms apply to any redistribution and
modification of the affected modules.
```

4.15 ZLIB DATA COMPRESSION LIBRARY

```
ZLIB DATA COMPRESSION LIBRARY

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  Jean-loup Gailly        Mark Adler
  jloup@gzip.org          madler@alumni.caltech.edu

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include third-party code.

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in the file ChangeLog history information documenting your changes. Please
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```

Chapter 5

Technical Support

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